

# Scaling Laws

DATA 443/643 · Week 7, note 7.1

Week 7

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2025 L9 (Kaplan curves; GPT-3 tokens vs Chinchilla). Original slides; that course is not copied.

# Lecture plan

- 1 Loss as a power of compute
- 2 A numeric on  $L(C)$
- 3 Chinchilla: tokens with parameters
- 4 What a law does not tell you

Key goal: write  $N$  and  $D$ , not only  $7B$ .  $L$  is pretraining CE, not chat quality.

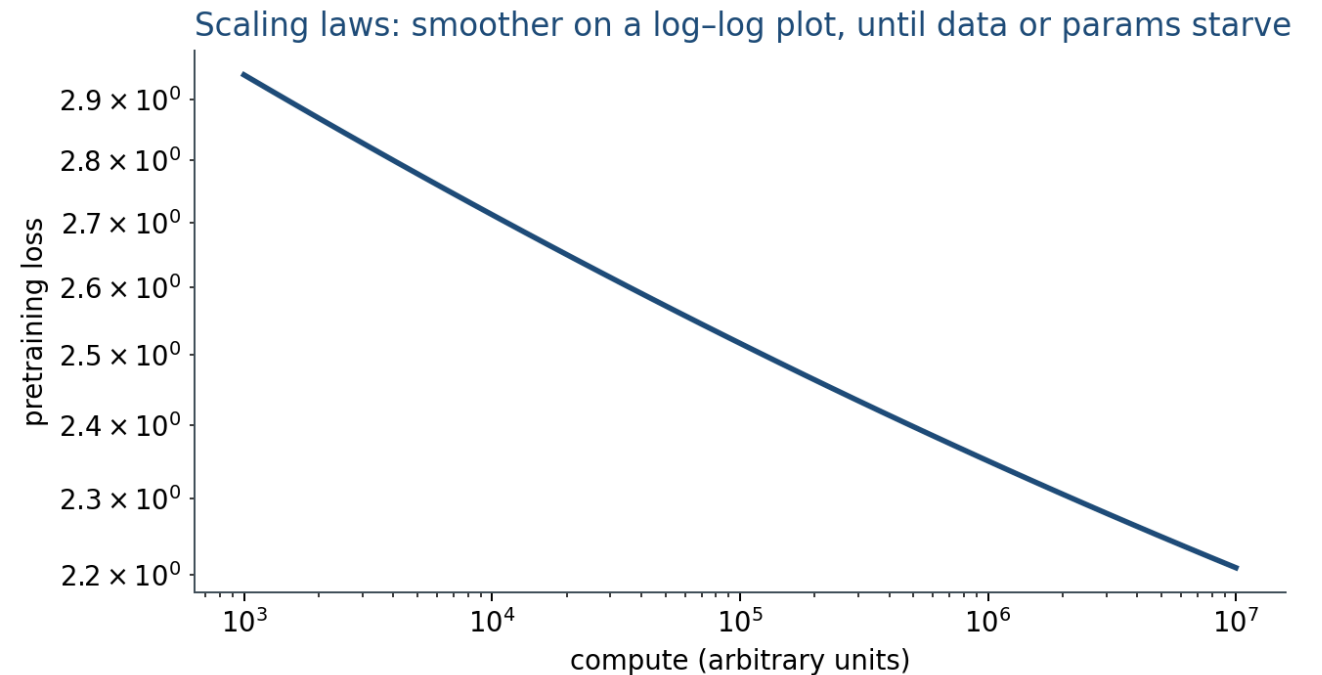
# First time: what / why / formula

- What / why. Kaplan: loss vs compute (and  $N, D$ ) looks like a power law plus a floor. Chinchilla: at fixed FLOPs, grow tokens and parameters together ( $\sim 20$  tokens/parameter in that paper). “We trained a 7B” is not a complete sentence. You need the token budget and whether you were compute-optimal.
- Architecture. Not a new net. A log–log plot and a recipe  $(N, D, C)$ .
- Formula.  $L(C) \approx aC^{-b} + L_\infty$ . Training FLOPs cartoon:  $\approx 6ND$ .
- Tradeoffs. + Planning a run. –  $L$  is not chat quality; data mixture and alignment sit off the curve; four toy points are not a law.

How. Fit  $L \approx aC^{-b} + L_\infty$  (Lab 7 algebra). Then do Chinchilla arithmetic:  $7B \times 20 = 140B$  tokens.

# A smoother log-log curve

- $L(C) \approx aC^{-b} + L_{\infty}$ .
- Extra FLOPs still buy loss, with diminishing returns.
- Lab 7 fits  $a$ ,  $b$ ,  $c$  on toy points. The paper's range is many decades.



# One hundred times the compute

$$L = 2 C^{-0.15} + 1.4$$

$$C = 10^4: L \approx 1.90.$$

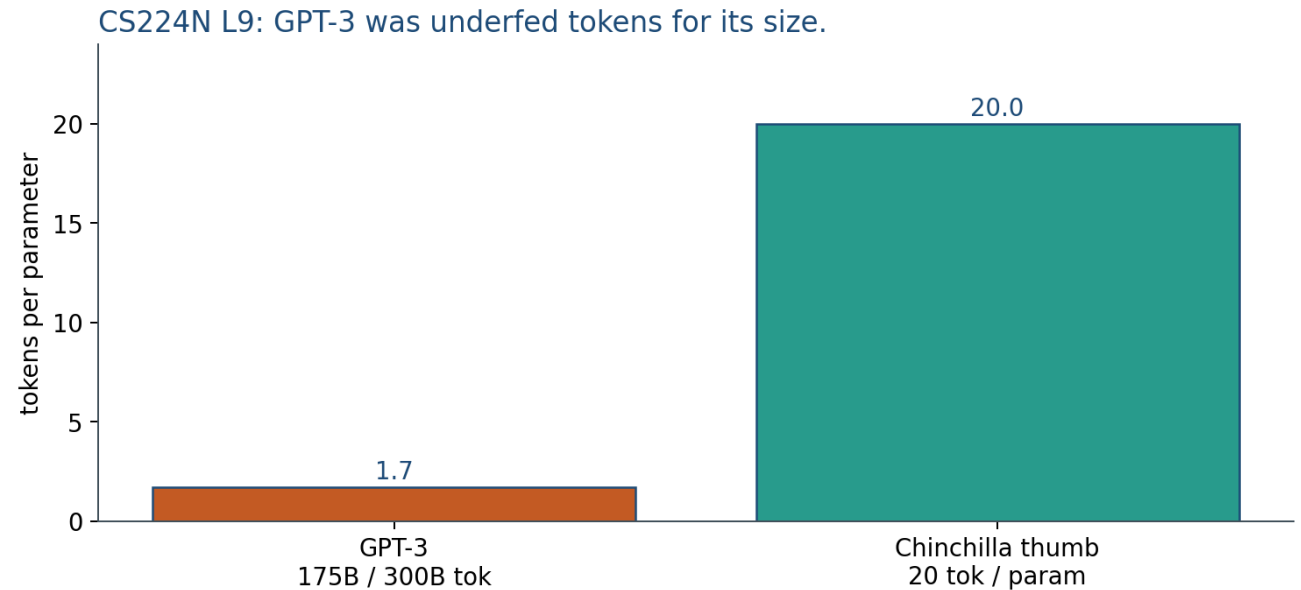
$$C = 10^6: L \approx 1.65. \text{ A}$$

100× jump bought 0.25 nats.

If you already sit on  $L_\infty$ , extra FLOPs buy almost nothing.

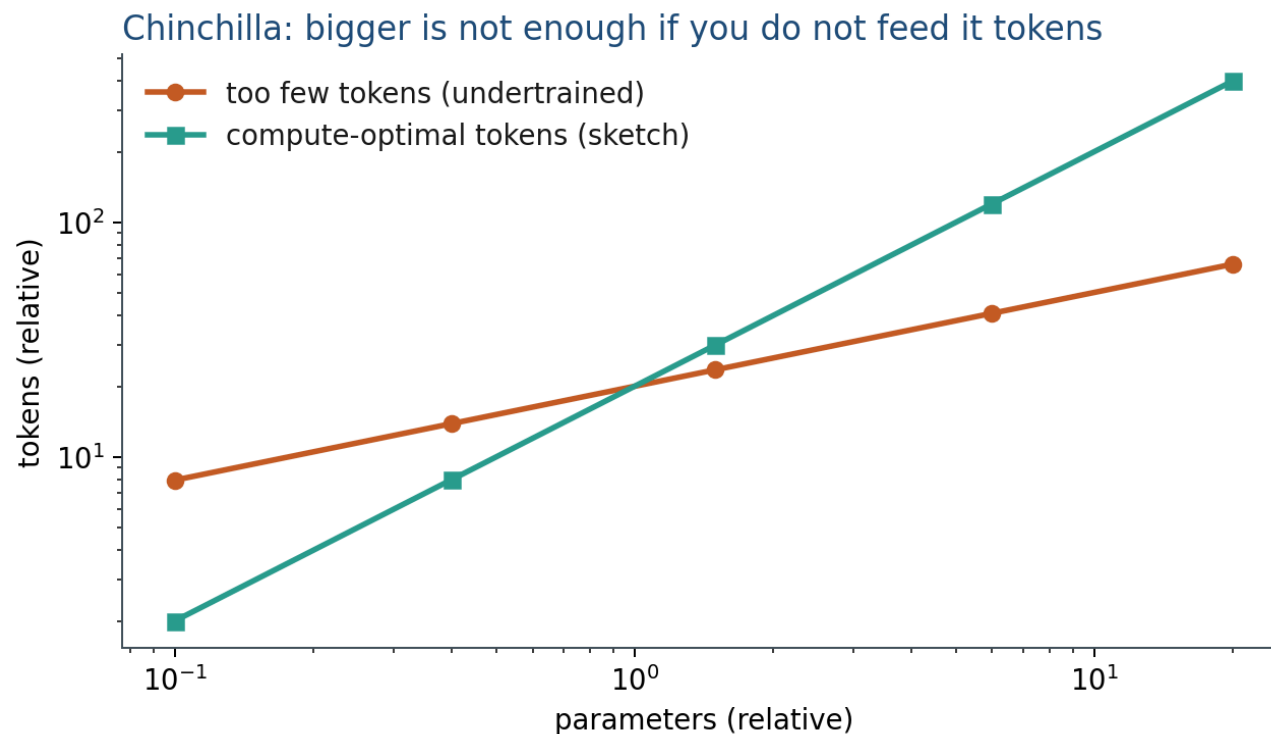
# GPT-3 was underfed

- GPT-3: 175B parameters, 300B tokens.
- That is about 1.7 tokens per parameter, not 20.
- A 70B trained with enough tokens beat larger starved nets.



# Same FLOPs, different (N, D)

- Training FLOPs scale like  $6ND$ .
- A:  $4N_0$  params,  $D_0 / 4$  tokens. B:  $N_0$ ,  $D_0$ .
- Chinchilla picks the balanced pair, not the wide starved net.



# What to take from this lecture

## Write D

7B × 20 tokens/param wants 140B tokens.

## L is not chat

Alignment, mixture, and eval sit off the curve.

## Lab 7

Fit a, b, c. Four noisy points are not a law.